

HL-CMR: Hypergraph Learning for Cross-Modal Retrieval

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Abstract

Cross-modal retrieval is a fundamental task in multimedia understanding, aimed at querying samples with similar semantics in one modality (e.g., text) using another modality (e.g., image). Existing methods merely focus on point-to-point comparisons between individual samples, while overlooking the widely present many-to-many structural relationships in real-world scenarios. However, the many-to-many relationships formed by multiple samples sharing similar semantics are crucial for effectively achieving semantic alignment and accurately constructing shared semantic representations. To address this, we propose a novel hypergraph-based cross-modal retrieval approach, which explicitly establishes many-to-many associations between multiple samples using a label-driven hypergraph construction mechanism, combined with differentiated hyperedge weighting. Additionally, to avoid the limitation of information interaction direction imposed by traditional unidirectional cross-attention mechanisms, we design a bidirectional cross-attention structure, with image and text as separate query sources, to achieve symmetric semantic enhancement between modalities. The resulting joint image-text representations are then mapped as hypergraph vertices, further enhancing the model's ability to align cross-modal semantics. Since constructing a global hypergraph on a large-scale sample set would incur high computational cost, we introduce global label co-occurrence frequency to supervise the batch-level hypergraph construction, enhancing the local graph's ability to capture global semantics. Experimental results show that our model outperforms existing state-of-the-art methods on three benchmark cross-modal retrieval datasets.

CCS Concepts

• Information systems → Information retrieval;

Keywords

Cross-modal retrieval, Hypergraph Learning, Multi-Label Scenarios.

ACM Reference Format:

Guohui Ding, Jing Li, Yimin Xu, and Rui Zhou. 2026. HL-CMR: Hypergraph Learning for Cross-Modal Retrieval. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3774904.3792526>

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ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792526>

1 Introduction

With the widespread application of multimedia data on social networks, the volume of data in different modalities such as images, text, video, and audio has grown explosively. Traditional single-modal retrieval methods can no longer meet the practical needs of increasingly diverse and heterogeneous information [14, 46, 47]. Consequently, cross-modal retrieval, as a key task in multimodal understanding, has attracted extensive attention. Its goal is to retrieve data from one modality using another modality by matching their semantics, thereby achieving semantic-level alignment.

Most existing methods remain at the level of point-to-point modeling between sample pairs (as shown in Figure 1(a)(b)), neglecting the richer semantic structural relationships among multiple samples [12, 45, 57]. In real-world applications such as social media and image-text recommendation scenarios, data often exhibits multi-label characteristics—i.e., a single sample may be associated with multiple semantic labels simultaneously, forming a typical multi-label scenario [58, 59]. In such cases, labels not only serve as semantic annotations for samples but can also act as latent semantic bridges, connecting multiple samples to form a group-level semantic structure, manifested as many-to-many relationships among samples (as illustrated in Figure 1(c)). The shared characteristics and similarities among multiple samples implied by these relationships can help align samples that share the same semantics across different modalities, thereby enabling more accurate semantic matching between images and texts. However, the semantic importance of the same label may vary significantly across different samples: for some, a shared label represents core semantics, while for others, it may only serve as background information. Modeling the relationships of samples with shared labels indiscriminately may introduce semantic bias and noise. Thus, we face the following challenge: How can we build a refined model of the many-to-many semantic relationships among samples in a multi-label scenario to overcome the limitations of traditional point-to-point modeling?

To address the above challenge, we propose, for the first time, a semantically weighted hypergraph structure that explicitly captures many-to-many relationships among samples by leveraging the natural high-order modeling capabilities of hypergraphs [12, 23]. Specifically, we use labels with high-level semantic information as hyperedges, as the connections formed by shared labels can clearly reflect semantic similarities between samples. Given that frequently co-occurring labels are often semantically related, we introduce global label co-occurrence frequency to measure the relevance between a shared label and the other labels of a sample. This relevance serves as an estimate of the semantic importance of the shared label to the sample, and is used as a differentiated weight between the sample and the corresponding hyperedge. This enhances both the

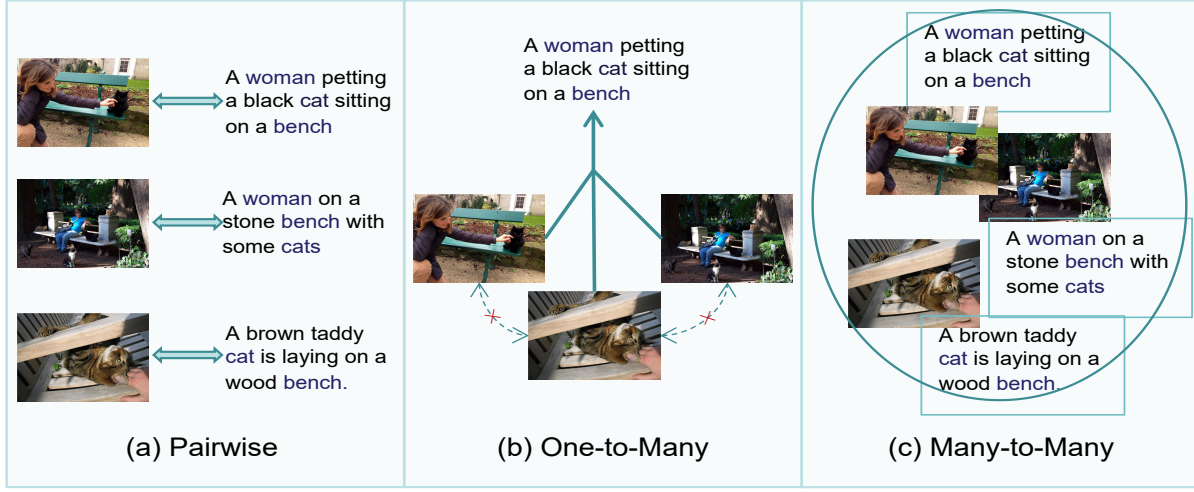


Figure 1: Comparison of semantic alignment strategies in cross-modal retrieval. (a) Pairwise: Each image-text pair is treated independently, ignoring other related samples. (b) One-to-Many: A single text is aligned with multiple images, but global semantic connections are still missing. (c) Many-to-Many: Multiple images and texts are grouped by shared semantics, enabling better modeling of high-order relationships.

semantic selectivity and robustness of information propagation within the hypergraph.

To construct hypergraph nodes, existing methods based on uni-modal features are insufficient for comprehensively capturing the semantic relationships between two modalities [33, 40, 57]. Therefore, we adopt a joint visual-textual feature representation to construct the hypergraph nodes. Meanwhile, to avoid the directional limitations in information interaction caused by traditional unidirectional cross-attention mechanisms [4, 22, 28, 48], we design a bidirectional cross-attention mechanism. Specifically, we incorporate a two-layer cross-attention structure, where image and text features query each other in two successive rounds. This enables the extraction of relevant information from each modality, achieving bidirectional semantic alignment and symmetric cross-modal interaction. As a result, the fused visual-textual features are mapped into hypergraph nodes with enhanced semantic integration and accuracy.

It is well known that constructing a global hypergraph over a large-scale sample set incurs extremely high computational overhead [6, 18]. To address this, we propose a batch-wise hypergraph construction method guided by global label co-occurrence frequency. By leveraging the global co-occurrence information as supervision, our approach ensures that the hypergraph within each batch can truly reflect the global semantic structure of the dataset, while significantly alleviating the computational cost associated with global hypergraph construction. The main contributions of our work are as follows:

- We propose a semantically weighted hypergraph modeling approach that explicitly captures many-to-many (high-order) relationships among samples in multi-label scenarios, effectively enhancing sample alignment capability.
- We design a bidirectional cross-attention module to fuse image and text modalities from heterogeneous to homogeneous

representations, thereby improving cross-modal understanding.

- We introduce a local-global collaborative graph construction mechanism, which, based on preserving global semantic priors, significantly reduces the model’s memory consumption and training cost.
- We conduct extensive empirical studies on three widely-used cross-modal retrieval benchmarks, and the results demonstrate that our method achieves significant performance improvements, outperforming various existing baselines.

The rest of the paper is organized as follows: In Section 2, we review related work; Section 3 presents our methodology; In Section 4, we describe the experiments and results; Section 5 concludes the paper.

2 Related Work

Traditional cross-modal retrieval methods primarily project heterogeneous modalities into a shared representation space through linear projections to enable similarity measurement [26, 31, 42]. Unsupervised methods such as CCA [13], RCCA [51], PLS [8], CMFH [9] and STMH [7] focus on structural modeling but often fall short in capturing semantic information. Supervised methods, including SePH [25], JRL [53], GMA [36] and SRLCH [37], incorporate label supervision to enhance modality alignment. However, most of these approaches rely on shallow or kernel-based models, which struggle to capture complex nonlinear relationships. With the development of deep learning, a growing number of deep cross-modal methods have been proposed. For instance, DCMH [17] introduces a negative log-likelihood loss for hashing, PRDH [50] strengthens intra-modal constraints, DSCMR [56] leverages modality-invariant loss and weight-sharing strategies to enhance performance, and MLCAH [29] utilizes attention mechanisms to improve robustness. DAGNN [33] and ALGCN [32] embed label information into graph

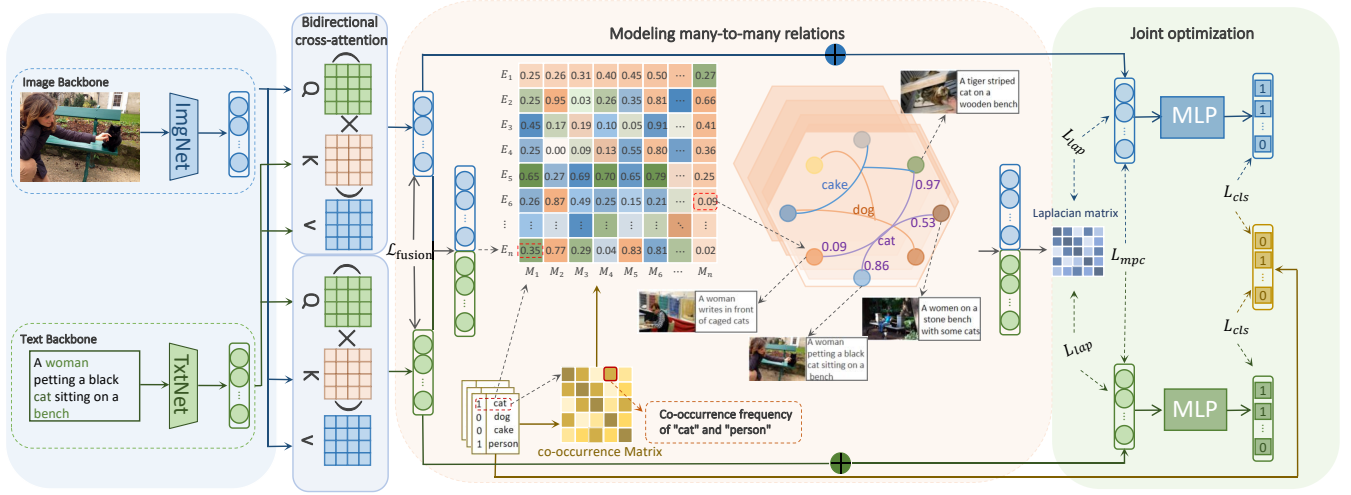


Figure 2: Overall framework of our proposed method.

neural networks, while DCPH [41] converts labels into proxy hash codes.

Despite these advancements in representation learning, structural information in cross-modal retrieval is often overlooked. Graph Neural Networks (GNNs), with their strong structural modeling capability, have been increasingly applied to cross-modal tasks. Models such as GAT [43] and RGAT [3] assign attention weights dynamically to capture complex dependencies. GCN-based [19] approaches like GCMLH [38], GCDH [1] and ALGCN [32] learn label semantics; AGCH [55] and AGCN [10] jointly model graph structures across image and text modalities; MMRG [52] integrates visual and textual relational graphs to enhance semantic expressiveness. Some studies also attempt to model semantic similarities between samples. Methods like GCH [49], WGSCN [54], and GCSCN [35] usually assess similarity based on shared labels, while HMAH [39] and PGCH [2] further consider the number of shared labels to model inter-sample relationships. However, these methods typically overlook the many-to-many structures among samples formed through labels and their interdependencies, making it difficult to effectively model semantic aggregation and thereby impairing the accurate measurement of semantic relatedness between samples.

To model many-to-many relationships, we introduce a hypergraph structure where labels are treated as hyperedges and fused image-text features as nodes. Hyperedge weights are computed based on global label co-occurrence, enabling more accurate sample-label association and ensuring consistency between cross-modal representations and the label space, thereby improving semantic alignment in retrieval.

3 Method

3.1 Problem Definition

Assume we have a multi-label dataset $\mathcal{O} = \{o_i\}_{i=1}^N$, where o_i denotes the i -th sample. Each sample consists of an image, text, and a corresponding label vector, i.e., $o_i = \{v_i, t_i, l_i\}$, where v_i and t_i represent the input image and text of the i -th sample, respectively.

$l_i \in \{0, 1\}^C$ is the label vector, where C is the total number of categories. Inspired by multi-label classification tasks [30], we assume that each sample can belong to multiple categories. If the i -th sample is assigned to the c -th category, then $y_i^c = 1$. The core of cross-modal retrieval is to establish semantic associations between different modalities, enabling efficient retrieval of semantically relevant content in one modality using data from another modality.

3.2 Model Overview

As illustrated in Figure 2, our framework comprises two key components: a Bidirectional Cross-Attention module and a many-to-many Relation Modeling module. In the Cross-Attention module, image and text features alternately serve as queries across two attention layers, enabling deep modality interaction and generating fused representations. In the Relation Modeling module, labels are treated as hyperedges and fused multimodal features as vertices. Hyperedge weights are computed based on label co-occurrence frequencies, capturing the associations between samples and labels at a more fine-grained semantic level. This mechanism enables fine-grained modeling of many-to-many relationships among samples. Additionally, we design a set of tailored loss functions to enable end-to-end joint training of the entire framework.

3.3 Bidirectional Cross-Attention Module

For the i -th input sample, we first feed the image v and text t into the encoders to extract the initial features f_v and f_t , which can be formulated as:

$$\begin{cases} f_v = \text{ImgNet}(v) \\ f_t = \text{TxtNet}(t) \end{cases} \quad (1)$$

where $\text{ImgNet}(\cdot)$ and $\text{TxtNet}(\cdot)$ denote the visual and textual feature extraction networks, respectively. where $f_v \in \mathbb{R}^{d_v}$ and $f_t \in \mathbb{R}^{d_t}$ represent the visual and textual features respectively, with d_v and d_t denoting their feature dimensions.

To more effectively capture the intercorrelations among heterogeneous modalities, the framework incorporates two independent

cross-attention layers. In particular, f_v and f_t are designated as the query vectors of the respective cross-attention layers, enabling the extraction of image-guided textual features and text-guided visual features. The input vectors for **Query**, **Key**, and **Value** are defined as follows:

$$\begin{cases} q_v = f_v W_1^Q, & k_t = f_t W_1^K, & v_t = f_t W_1^V, \\ q_t = f_t W_2^Q, & k_v = f_v W_2^K, & v_v = f_v W_2^V. \end{cases} \quad (2)$$

Here, q , k , and v denote the query, key, and value vectors, respectively. The matrices $W_1^Q, W_1^K, W_1^V \in \mathbb{R}^{d \times d}$ are learnable linear projections associated with the i -th attention layer, where $d = d_v = d_t$. Accordingly, the image-guided textual feature $x_{t \rightarrow v}$ and text-guided visual features $x_{v \rightarrow t}$ can be derived by self-attention as,

$$\begin{cases} x_{t \rightarrow v} = \text{SAtt}(q_v, k_t, v_t) = \sum_{j=1}^m \text{softmax} \left(\frac{q_v k_{t,j}^\top}{\sqrt{d_k}} \right) v_{t,j}, \\ x_{v \rightarrow t} = \text{SAtt}(q_t, k_v, v_v) = \sum_{j=1}^m \text{softmax} \left(\frac{q_t k_{v,j}^\top}{\sqrt{d_k}} \right) v_{v,j}, \end{cases} \quad (3)$$

where m denotes the number of tokens in either the image or text sequence, while $k_{\cdot,j}$ and $v_{\cdot,j}$ represent the key and value vectors of the j -th text token, respectively. The function SAtt refers to the standard self-attention mechanism, where K and V are the key and value matrices involved in the computation.

To further enhance the extraction of text-guided visual features, a Feed-Forward Network (FFN) is incorporated, yielding multimodal representations that can be written as:

$$y_{t \rightarrow v} = RS_2(RS_1(x_{t \rightarrow v})), \quad (4)$$

where RS_1 and RS_2 denote two consecutive residual layers, which can be formulated as:

$$RS_1(\alpha) = \text{Norm}(\text{Drop}(\alpha) + \alpha), \quad (5)$$

$$RS_2(\alpha) = \text{Norm}(\text{Drop}(\text{FFN}(\alpha)) + \alpha), \quad (6)$$

Here, **Norm** and **Drop** represent the normalization and dropout operations, respectively. The resulting $y_{t \rightarrow v}$ is the enhanced text-guided visual feature representation, and, by symmetry, the enhanced image-guided textual features are obtained through the analogous procedure illustrated in Eq 4.

To ensure semantic consistency among samples from the same modality and class, we introduce the modality contrastive loss, defined as:

$$\begin{aligned} \mathcal{L}_{\text{fusion}}^I &= - \sum_{i=1}^N \log \frac{\exp(\langle y_{t \rightarrow v}^i, y_{v \rightarrow t}^i \rangle / \tau)}{\sum_{j=1}^N \exp(\langle y_{t \rightarrow v}^i, y_{v \rightarrow t}^j \rangle / \tau)}, \\ \mathcal{L}_{\text{fusion}}^T &= - \sum_{i=1}^N \log \frac{\exp(\langle y_{v \rightarrow t}^i, y_{t \rightarrow v}^i \rangle / \tau)}{\sum_{j=1}^N \exp(\langle y_{v \rightarrow t}^i, y_{t \rightarrow v}^j \rangle / \tau)}. \end{aligned} \quad (7)$$

Here, N is the number of visual or textual samples, τ the temperature parameter, and $\langle \cdot, \cdot \rangle$ denotes the similarity function. The multi-modal contrastive loss is the summation of Eq 7 as:

$$\mathcal{L}_{\text{fusion}} = \mathcal{L}_{\text{fusion}}^I + \mathcal{L}_{\text{fusion}}^T \quad (8)$$

3.4 Many-to-Many Relationship Modeling Module

3.4.1 Construction of High-order Relationship Modeling Module. The High-order Relation Modeling module enables the learning of fine-grained cross-modal relationships by leveraging the high-order connectivity represented through hypergraph structures. Specifically, given a label matrix $M \in \mathbb{R}^{N \times L}$, where N is the number of samples and L is the total number of labels, we derive the label co-occurrence matrix $C \in \mathbb{R}^{L \times L}$ as follows:

$$C = \frac{M^\top M}{N}. \quad (9)$$

Based on the co-occurrence matrix, we construct a hypergraph connection matrix $H \in \mathbb{R}^{N \times L}$, where each column represents a hyperedge corresponding to a label, and each row denotes a sample node. The entry H_{ij} indicates the connection strength between i -th sample and j -th label. Its value is obtained by summing the co-occurrence frequencies of label j with other labels present in sample i , which can be defined as:

$$H_{ij} = \begin{cases} \sum_{\substack{k=1 \\ k \neq j}}^L \mathbf{1}_{M_{ik}=1} \cdot C_{jk}, & \text{if } M_{ij} = 1, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

Here, C_{ij} represents the co-occurrence strength between labels i and j , reflecting how frequently the two labels appear together across all samples. $\mathbf{1}_{M_{ik}=1}$ is an indicator function that equals 1 if sample i also possesses label k , and 0 otherwise. Hence, sample i is connected to hyperedge j only if it holds label j , and the connection weight is further influenced by other co-occurring labels.

To propagate semantic information in the hypergraph, we compute the normalized hypergraph propagation matrix $G \in \mathbb{R}^{N \times N}$ from matrix H . This matrix enables structured semantic information propagation, allowing each sample to aggregate related features from other samples within the same hyperedge:

$$G = D_v^{-\frac{1}{2}} H W D_e^{-1} H^\top D_v^{-\frac{1}{2}}, \quad (11)$$

where $D_v \in \mathbb{R}^{N \times N}$ is the diagonal degree matrix of the sample nodes, $D_e \in \mathbb{R}^{L \times L}$ is the diagonal degree matrix of the hyperedges, $W \in \mathbb{R}^{L \times L}$ is the learnable hyperedge weight matrix.

3.4.2 Overall Pipeline of High-order Relationship Modeling Module. We concatenate the image-text fusion features obtained from the bidirectional cross-attention module as the input to the High-order Relation Modeling (HRM) module, which is denoted as:

$$Z \leftarrow \text{concat}[Y_{v \rightarrow t}, Y_{t \rightarrow v}]. \quad (12)$$

We utilize a two-layer Hypergraph Convolutional Network (HGNN) to extract high-order relationships. Specifically, the hypergraph convolution operation leverages the hypergraph propagation matrix G and hypergraph connection matrix H to transmit and update features, which is defined as:

$$\text{HGConv}(Z, H) = GZH. \quad (13)$$

The final output of the HGNN is then represented as:

$$Z^* = \text{HGNN}(Z, H) = \text{Norm}(\text{HGConv}_2(\text{ReLU}(\text{HGConv}_1(Z, H)))) \quad (14)$$

To integrate the original features with the structure-aware representations, the enhanced visual-textual representation E is obtained through a gated fusion of the propagated and original features:

$$E = \lambda Z^* + (1 - \lambda)Z, \quad (15)$$

where λ is the fusion coefficient controlling the contribution of enhanced hypergraph features.

3.5 Joint Optimization

To improve high-order semantic structure modeling and enhance representation consistency among semantically related samples, we introduce a hypergraph loss consisting of two components: a multi-positive contrastive loss and a Laplacian smoothness loss. The former is defined as:

$$\mathcal{L}_{\text{mpc}} = - \sum_{i=1}^N \frac{\sum_{j \in g(i)} \exp\left(\frac{\langle e_i, e_j \rangle}{\tau}\right)}{\sum_{k=1}^N \exp\left(\frac{\langle e_i, e_k \rangle}{\tau}\right)}. \quad (16)$$

Here, e_i denotes the enhanced visual-textual representation of the i -th sample, N is the total number of samples, and $g(i)$ denotes the set of indices corresponding to samples that share at least one label with the i -th sample. The term $\langle e_i, e_j \rangle$ measures the similarity between e_i and e_j , while τ is a temperature parameter.

To ensure smoothness and structural consistency of neighboring samples in the feature space, we incorporate a Laplacian smoothness loss, defined as:

$$\mathcal{L}_{\text{lap}} = \frac{1}{N} \cdot \text{Tr}(E^T L E), \quad (17)$$

where $L = D - S$ is the Laplacian matrix constructed from sample affinity based on shared labels. S is defined as:

$$S_{ij} = \frac{M_i \cdot M_j}{\|M_i\|_1}, \quad (18)$$

which quantifies the degree of label overlap between sample i and sample j . D is a diagonal matrix whose diagonal entries are the row sums of S , i.e.,

$$D_{ii} = \sum_j S_{ij}. \quad (19)$$

Combining both losses, the total hypergraph loss is defined as:

$$\mathcal{L}_{\text{HGNN}} = \alpha \mathcal{L}_{\text{mpc}} + \beta \mathcal{L}_{\text{lap}} \quad (20)$$

To support subsequent classification and retrieval tasks, we split the enhanced visual-textual representation E into visual and textual modalities according to their corresponding dimensionalities:

$$[E_v, E_t] \leftarrow \text{split}(E, [d, d]). \quad (21)$$

To enhance semantic discrimination, we project the enhanced visual and textual features into their corresponding semantic spaces:

$$\begin{cases} P_v = W_2^v \cdot \text{ReLU}(W_1^v \cdot E_v + b_1) + b_2, \\ P_t = W_2^t \cdot \text{ReLU}(W_1^t \cdot E_t + b_1) + b_2, \end{cases} \quad (22)$$

where $W_1^v, W_1^t \in \mathbb{R}^{d_n \times d}$ first project the original features into a hidden dimension d_n . Then, $W_2^v, W_2^t \in \mathbb{R}^{L \times d_n}$ project the hidden features into a dimensional space consistent with the multi-label prediction task for subsequent classification. b_1 and b_2 denote the corresponding bias terms, and $\text{ReLU}(\cdot)$ is the activation function.

Algorithm 1 Optimizing process of the proposed HL-CMR

Input: N Image-text training pairs; batch size M ; The number of epochs Q ; learning rate η ; The hyper-parameters α, β and λ .

Parameter: Bidirectional Cross-Attention: μ^I and μ^T ; High-order Relationship Modeling: δ ; Classification MLP: ϵ^I and ϵ^T .

Output: Learned multimodal representation E .

repeat

for $i \in [N/M]$ **do**

Randomly select M training instance pairs;

Fusion F_v, F_t and by eq (2) and (3);

Calculate Contrastive loss by eq (7);

Construct the Hypergraph G by eq (10) and (11);

Generate Enhance feature E by eq (14) and (15);

Calculate Hypergraph loss by eq (19);

Prediction multi-label by eq (21);

Update all parameters by eq (24);

end for

until maximum epoch Q is reached

return Learned multimodal representation E .

To enhance semantic discrimination, we perform multi-label classification on image and text features separately, using Binary Cross-Entropy (BCE) loss to guide each modality in accurately predicting label semantics. The loss is defined as:

$$\mathcal{L}_{\text{cls}} = \frac{1}{2} (BCE(P_v, M) + BCE(P_t, M)), \quad (23)$$

where P_v and P_t represent the image and text embeddings projected via MLP, and M denotes the ground-truth label matrix.

The final loss function combines the above components, and the overall optimization objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{fusion}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{HGNN}}. \quad (24)$$

The whole alternating learning algorithm for the proposed HL-CMR is briefly outlined in Algorithm 1.

Table 1: Statistical Information of Experimental Datasets

Dataset	Total	Query/Retrieval/Train	Category
MIR Flickr	20,015	2,000/18,015/5,000	24
NUS-WIDE	190,421	2,000/188,421/5,000	21
MS COCO	122,218	2,000/120,218/5,000	80

4 Experiments

4.1 Datasets

To verify the effectiveness of the proposed model HL-CMR, we conduct experiments on three popular benchmark datasets for cross-modal retrieval with shown in Table 1.

MIRFlickr-25K[15]: The MIRFlickr dataset contains 25,000 image-text pairs across 24 categories, with each image annotated by multiple tags. We use 20,015 pairs for experiments, including 2,000 for querying and 18,015 for retrieval. From the retrieval set, 5,000 samples are randomly selected for training.

NUS-WIDE[5]: The NUS-WIDE dataset contains 269,648 image-text pairs across 81 categories. We use 190,421 pairs from the top 21 most frequent categories, with 2,000 as the query set and 188,421

Table 2: Comparison of mAP scores on three datasets.

Method	NUS-WIDE			MIRFlickr-25K			MS-COCO		
	I2T	T2I	Average	I2T	T2I	Average	I2T	T2I	Average
CFA [20]	0.354	0.361	0.357	0.580	0.548	0.564	0.340	0.397	0.369
JRL [53]	0.427	0.361	0.394	0.589	0.554	0.571	0.388	0.364	0.376
CDTH [21]	0.670	0.691	0.681	0.745	0.746	0.746	0.657	0.657	0.657
ACMR [44]	0.684	0.675	0.680	0.736	0.748	0.742	0.706	0.708	0.707
DNpH [34]	0.707	0.714	0.711	0.859	0.836	0.848	0.690	0.693	0.692
DSPH [16]	0.716	0.730	0.723	0.854	0.829	0.842	0.769	0.771	0.770
MS ² GAH [11]	0.727	0.749	0.738	0.780	0.797	0.788	0.824	0.819	0.821
GCDH [1]	0.736	0.751	0.743	0.783	0.804	0.793	0.827	0.823	0.825
DCH-SCR [27]	0.742	0.764	0.753	0.853	0.855	0.854	0.617	0.573	0.595
MGSSL [58]	0.774	0.779	0.777	0.864	0.866	0.865	0.858	0.860	0.859
HL-CMR	0.823	0.836	0.829	0.890	0.900	0.895	0.869	0.890	0.879

for retrieval. Among them, 5,000 samples are randomly selected for training.

MS COCO[24]: The MS COCO dataset includes 123,287 image-text pairs across 80 categories. After filtering out unlabeled samples, 122,218 pairs are retained, with 2,000 used for querying, 120,218 for retrieval, and 5,000 randomly selected for training.

4.2 Baselines and Evaluation Metrics

We compare our proposed method with 10 cross-modal retrieval baselines, including two traditional methods CFA [20] and JRL [53], as well as 8 deep learning-based approaches: CDTH [21], ACMR [44], DNpH [34], DSPH [16], MS²GAH [11], GCDH [1], DCH-SCR [27], and MGSSL [58].

Following previous studies, we adopt the widely used mean Average Precision (mAP) to evaluate the retrieval performance of all methods. Specifically, for a given query, we first compute its Average Precision (AP), and the mAP is calculated by averaging the AP values over all queries. In addition, we plot the Precision-Recall (PR) curves to further assess the performance of all the methods.

4.3 Implementation Details

For the image modality, we adopt a convolutional neural network CNN-F pretrained on the ImageNet dataset as the image feature extractor. For the text modality, we design an MLP with 512 output nodes to map the original bag-of-words vectors into continuous-valued features.

we construct a network architecture consisting of three main components: The Bidirectional Cross-Attention module is defined as a two-layer feed-forward network: $(512 \rightarrow 256 \rightarrow \text{ReLU} \rightarrow 512 \rightarrow \text{Tanh})$ The many-to-many Relationship Modeling module performs hypergraph propagation on image and text embeddings to enhance feature representation. Its architecture is defined as: $(1024 \rightarrow \text{HGNN_conv}(1024) \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{HGNN_conv}(1024) \rightarrow \text{ReLU} \rightarrow \text{Linear}(1024) \rightarrow \text{L2_normalize})$ The MLPs used for projecting image and text features into the shared label space follow the structure: $(512 \rightarrow 256 \rightarrow \text{ReLU} \rightarrow \text{num_labels})$ where num_labels denotes the number of label categories in the dataset. we extract 512-dimensional image and text features (F^I, F^T) using modality-specific feature extraction networks. These are fused via a Bidirectional Cross-Attention Module

to generate a shared embedding, optimized with a contrastive loss function to align paired image-text samples and preserve cross-modal semantic consistency. Next, the many-to-many Relationship Modeling module further processes these fused features via a hypergraph propagation mechanism, where a hypergraph loss promotes semantic clustering and penalizes weakly related sample associations. We also adopt MLPs to project features into a shared label space, supervised by a multi-label classification loss to enhance semantic alignment and multi-label recognition.

The proposed model is implemented using the open-source PyTorch framework and trained on a server equipped with two NVIDIA TITAN X GPUs. The training hyperparameters are set as follows: learning rate = 10^{-4} , batch size = 128, and the model is optimized using the Adam optimizer.

4.4 Experiment Results

Table 2 presents the mAP scores of our model and ten baselines on three benchmark datasets for two tasks: I→T and T→I retrieval. Based on the experimental results, we draw the following conclusions:

Results show that deep cross-modal methods outperform traditional ones, confirming the advantages of end-to-end architectures in learning cross-modal representations. Our model outperforms all baseline methods across the three datasets and achieves substantial improvements on the NUS-WIDE and MIRFlickr-25K datasets. Specifically, compared with the second-best results, our model improves the average mAP scores by 5.2% on NUS-WIDE, 3.0% on MIRFlickr-25K, and 2.0% on MS-COCO. Additionally, Figure 3 shows the Precision-Recall curves, where our method consistently outperforms all baselines, further validating its effectiveness and superiority.

This improvement mainly stems from the fact that existing methods often overlook the multi-label scenario, where the relationships between samples are not simply one-to-one or one-to-many, but rather many-to-many group relationships formed by multiple samples sharing the same or similar semantics. Furthermore, we plot the Precision-Recall Curves on the three datasets, as shown in Figure 3. The results demonstrate that our method consistently achieves the best performance among all compared approaches, further validating the effectiveness and superiority of our method.

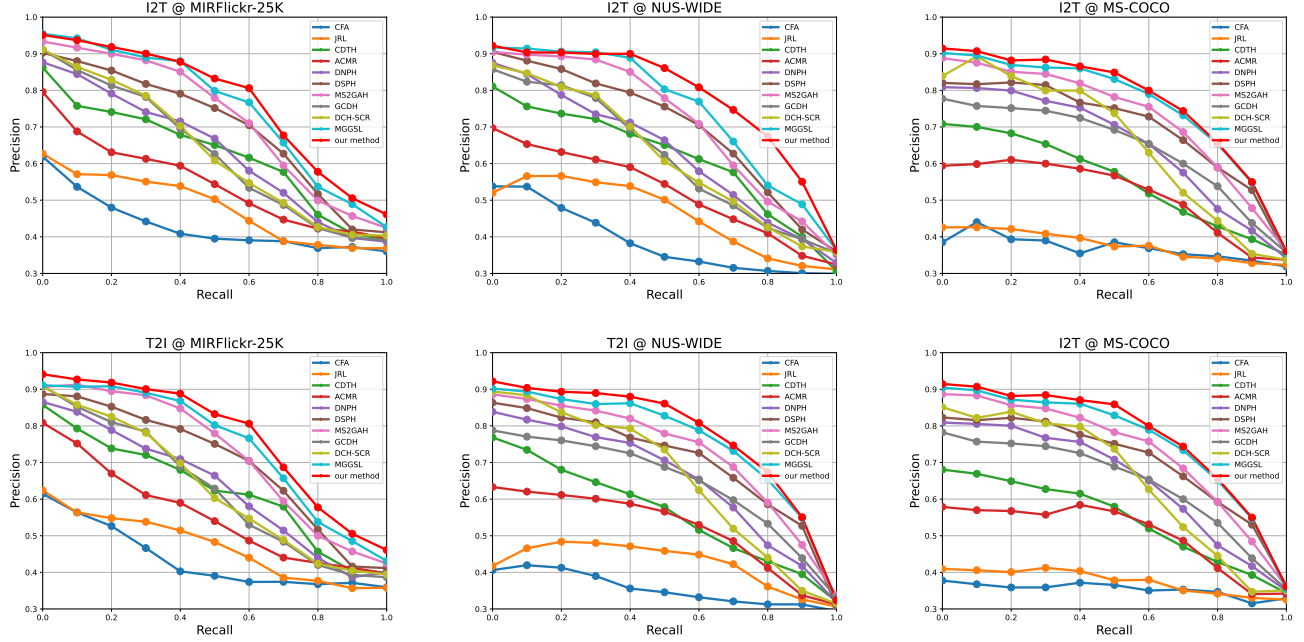


Figure 3: Precision-Recall Curves on the three datasets.

4.5 Ablation Study

To evaluate the effectiveness of the key components and modules of the proposed model, we design four variants for comparison: (1) **w/o 1**: removing the High-order Relationship Modeling module for hypergraph-based high-order relation modeling. (2) **w/o 2**: replacing the hypergraph neural network with a standard graph neural network. (3) **w/o 3**: removing the Bidirectional Cross-Attention module. (4) **w/o 4**: removing the differentiated weighting design in the hypergraph and adopting uniform information propagation across samples. Based on the results reported in Table 3, we draw the following analyses:

Removing the High-order Relationship Modeling module leads to a significant drop in retrieval performance. This demonstrates that high-order relation modeling plays a crucial role in capturing complex semantic relationships among samples. Without this module, the model’s ability to exploit global structural information is considerably weakened, thereby impairing its final discriminative performance. Replacing the hypergraph neural network with a conventional graph neural network leads to a significant performance drop, which is even more pronounced on complex datasets. When using a conventional graph structure, the model fails to fully capture the many-to-many relationships between samples, thereby limiting its performance across various tasks. In contrast, the hypergraph structure can effectively model complex relationships, enrich sample representations, and consequently achieve a substantial improvement in overall performance.

Removing the Bidirectional Cross-Attention module results in a substantial decrease in retrieval performance. Experimental results clearly show that, without this module, cross-modal alignment becomes insufficient, leading to large discrepancies between the two

retrieval tasks. By contrast, multimodal feature fusion enhances and enriches semantic representations, while transforming heterogeneous states into homogeneous ones, effectively mitigating the impact of the semantic gap on cross-modal retrieval. Removing the differentiated weighting design yields the most significant drop in retrieval performance. In this variant, a mean propagation strategy is adopted, which neglects the importance differences among nodes. Consequently, the model fails to highlight the influence of key nodes during information propagation and becomes more susceptible to noise. Differentiated weighting design is therefore essential for enhancing the model’s discriminative capability and semantic aggregation efficiency.

Table 3: The mAP scores of our method variants on three datasets.

Methods	NUS-WIDE		MIRFlickr-25K		MS-COCO	
	I2T	T2I	I2T	T2I	I2T	T2I
w/o 1	0.774	0.786	0.871	0.882	0.818	0.829
w/o 2	0.791	0.809	0.882	0.890	0.785	0.797
w/o 3	0.785	0.793	0.855	0.876	0.788	0.810
w/o 4	0.722	0.732	0.821	0.829	0.763	0.772
HL-CMR	0.823	0.836	0.890	0.900	0.869	0.890

4.6 Parameter Sensitivity

We analyzed the impact of the parameters α , β , and λ on the training process. The combination of α and β controls the feature space, bringing samples with shared labels closer and evenly distributing

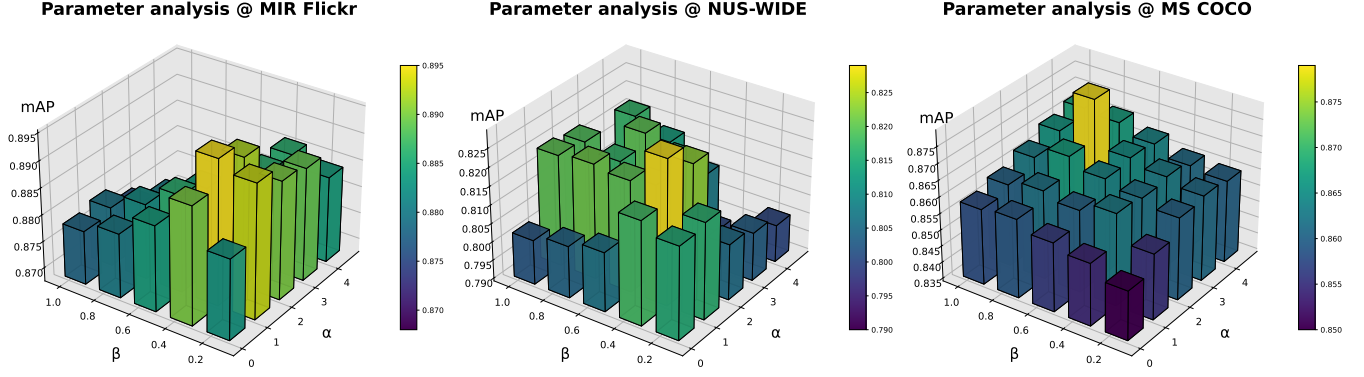


Figure 4: α, β Parameter sensitivity analysis on MIR Flickr, NUS-WIDE, and MS COCO datasets

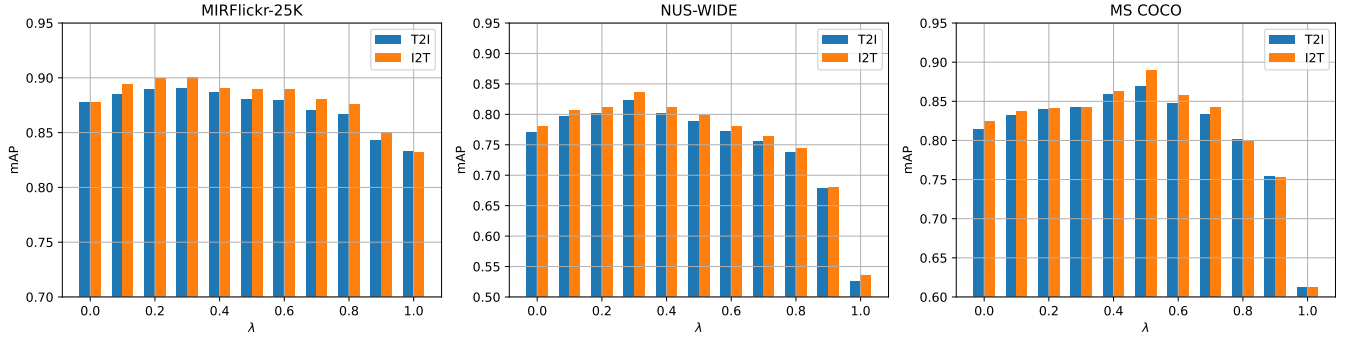


Figure 5: λ Parameter sensitivity analysis on MIR Flickr, NUS-WIDE, and MS COCO datasets

samples with overlapping labels, resulting in a semantically compact and structurally smooth representation that improves feature expressiveness for downstream retrieval tasks. We set the ranges of α and β as $\{0, 1, 2, 3, 4, 5\}$ and $\{0.2, 0.4, 0.6, 0.8, 1.0\}$, respectively, and evaluated the effect of varying one parameter (e.g., α) while fixing the other (e.g., β). The Figure 4 presents the mean average precision (mAP) results on two retrieval tasks across three datasets. When adjusting α and β , the best performance on the MIRFlickr-25K and NUS-WIDE datasets occurred at $\alpha = 1$ and $\beta = 0.3$, while the best performance on the MSCOCO dataset was at $\alpha = 3$ and $\beta = 0.7$. We set the range of λ to $[0, 1.0]$ with a step size of 0.1. The parameter λ controls the proportion between the fused features and the hypergraph-enhanced features, with the best results achieved at $\lambda = 0.3$ for MIRFlickr-25K and NUS-WIDE, and at $\lambda = 0.5$ for MSCOCO. The Figure 5 shows the mAP results on two retrieval tasks and three datasets. In the MIRFlickr-25K and NUS-WIDE datasets, the number of label categories is relatively small, and each sample has only a few labels. If the hypergraph weight is set too high, many weakly related samples are aggregated, leading to the introduction of false positives. In contrast, the MSCOCO dataset contains more categories, and its label co-occurrence matrix has a higher rank and denser semantic relationships. Increasing the hypergraph proportion helps the model better capture these dense and overlapping label structures, while a lower proportion may result in insufficient smoothing.

5 Conclusion

In this paper, we propose a novel model HL-CMR that integrates Bidirectional Cross-Attention and a semantically weighted hypergraph. By dynamically modeling many-to-many relationships through globally co-occurrence weighted hyper-edges. By leveraging Bidirectional Cross-Attention, the fused image-text features are used as hypergraph vertices. Our approach simultaneously addresses computational and memory bottlenecks in large-scale datasets via a local-batch-construction and global-optimization strategy, while significantly improving cross-modal retrieval accuracy. Extensive experiments on three popular cross-modal benchmark datasets demonstrate the superiority of HL-CMR.

6 Acknowledgments

This work was funded by the project ZR2022MF333 supported by Shandong Provincial Natural Science Foundation, the Key Laboratory of Computing Power Network and Information Security, Ministry of Education under Grant No. 2023ZD028, the Sci-Tech Innovation 2030—“New Generation Artificial Intelligence” Major Project (2018AAA0102100), the Natural Science Foundation of Liaoning Province (2021-MS-261), the Key Project of the Natural Science Foundation of Zhejiang Province (LZ22F020014), the Key R&D Sub-Project of the Ministry of Science and Technology (2021YFF0307505) and the Liaoning Provincial Project (XLYC2203002).

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